Name: ____________________________________

Name the condition that proves the quadrilateral is a parallelogram, or write NOT ENOUGH.

1. Both pairs of opposite sides are congruent.

2. One pair of opposite sides is parallel and the OTHER pair is congruent.

3. The diagonals bisect each other.

4. One pair of opposite sides is both parallel and congruent.

5. Both pairs of opposite angles are congruent.

Solve so the quadrilateral is a parallelogram.

6. Opposite sides measure $7x - 9$ and $4x + 15$, and the other pair measures $6y + 1$ and $9y - 20$. Find x and y.

7. Opposite angles measure $(13m - 11)°$ and $(9m + 25)°$. Find m and the angle.

8. The diagonals meet at T. One diagonal has halves $5k - 7$ and $3k + 9$; the other has halves $2k + 6$ and $4k - 10$. Find k.

Coordinates.

9. Show that $W(-2, 1)$, $X(2, 4)$, $Y(7, 4)$, $Z(3, 1)$ is a parallelogram using slopes.

10. Show the same thing a second way, using lengths (distance formula).